The first law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a consistent sign convention in an energy balance.

SUCCESS CRITERIA - I can...

1. State whether heat enters or leaves.
2. Distinguish work done by the system from work done on it.
3. Apply ΔU=Q-W with the stated signs.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A 0.3 kg liquid releases 100000 J while condensing at constant temperature. Find latent heat and signed Q for the sample.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Delta U = Q - W_{by}$.
Q is positive into the system; W_{by} is positive when the system does work on its surroundings.

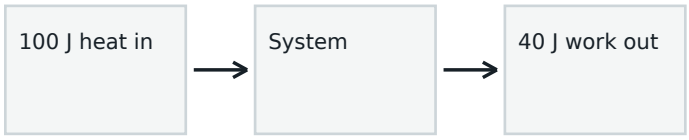
Worked reasoning: A system receives 90 J of heat and does 30 J of work on its surroundings. With ΔU=Q-W, internal energy increases by 60 J.

Your turn: Draw labeled arrows crossing a system boundary and mark the 60 J stored increase.

Marzano check - circle: ① ② ③

03 Represent and interpret

A system absorbs 100 J of heat and does 40 J of work. Draw the energy arrows and find its internal-energy change.



Marzano check - circle: ① ② ③

04 Solve and justify

A gas loses 50 J of heat while 60 J of work is done on it. Use the reference sign convention to calculate ΔU.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

Why does the phrase “work is 60 J” need a direction or sign convention?

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg cart at 12 m/s hits an identical cart at rest. Final velocities are 5 and 7 m/s. Check momentum and energy; classify the collision.

Marzano check - circle: 1 2 3

07 Exit check

A process hasΔU=-15 J and W_{by}=25 J. Find Q and say whether heat enters or leaves.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A system loses 20 J of heat and has 50 J of work done on it. Using work-by-system as positive, find Q, W and ΔU. Then reverse both transfers.

Marzano check - circle: 1 2 3

Read down the left column, then down the right. Use the open space to draw, calculate and explain.